

Favour O. Igwezeke

favour.igwezeke.249461@unn.edu.ng | [favourigwezeke.github.io](https://github.com/favourigwezeke) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [ORCID](#)

Education

University of Nigeria, Nsukka

Doctor of Pharmacy (PharmD)

Nsukka, Nigeria

Expected Feb. 2027

CGPA: 4.21/5.0 (Yr 1–5) · 4.58/5.0 (Yr 2–5)

Thesis: “Deep Learning–Based Behavioral and Anxiolytic Effects of the Methanol Stem Bark Extract of *Indigofera bracteolata* (DC.) in Wistar Rats”

Advisor: Assoc. Prof. Collins A. Onyeto, Dept. of Pharmacology & Toxicology, UNN

Research Interests

Biologically grounded representation learning; mechanistic and context-aware biomedical AI; uncertainty-aware reasoning and distribution shift in clinical and computational biology models; drug response prediction; perturbation modeling and transferability of biological representations.

Research Experience

SPARK (MICCAI-Endorsed Academy for African Medical Imaging Translation)

Remote

ML Research Fellow — Medical Imaging AI

Feb. 2026 – Present

- **MORPHA** — Tested whether morphological priors improve cross-acquisition transfer in malaria microscopy; on Ugandan field data, reduced the transfer drop by 30.8% and halved calibration error, with a content-free control isolating morphology. Accepted at the **MIRASOL Workshop @ MICCAI 2026**; forthcoming in *Springer LNCS*. [Paper]

CaresAI Research Group

Remote

Medical AI Researcher

Jan. 2026 – Present

Group Lead: Dr. Mary Adewunmi

- **Mechanism or Memorisation?** — Introduced **cross-endpoint relational support** (R_X), a training-relative diagnostic separating entity-level familiarity from relationship-level support in DTI prediction; showed that high- R_X interactions achieve consistently higher matched AP and lower adjusted loss across three representation families on 80,094 BindingDB interactions. Submitted to *ICBINB-BIO @ NeurIPS 2026*.
- **CURE** — Developed an uncertainty-aware Graph-RAG system for evidence-grounded multihop biomedical reasoning; used **selective claim-level evidence verification**, backtracking, and abstention to study reliability under incomplete evidence. Submitted to *ML4H 2026, Proceedings Track*.
- **CT-DEB’26** — Developed an ensemble ML framework integrating four domain-specific BERT embeddings to predict medication dosing errors across 29,000+ ClinicalTrials.gov protocols. Published at **CL4Health @ LREC 2026**. [Paper]

Dept. of Pharmacology & Toxicology, University of Nigeria, Nsukka

Nsukka, Nigeria

Undergraduate Researcher — Computational Toxicology & Neuropharmacology

Nov. 2024 – Present

Supervisor: Assoc. Prof. Collins A. Onyeto

- **CellStateGRN** (ongoing) — Formulating a **perturbation-anchored causal representation framework** to identify shared regulatory programs and sparse state-specific rewiring across compensating and proliferating hepatocyte states in liver repair; planned validation against synthetic ground truth and evaluation on interventions withheld from model fitting.
- **TrajRisk** — Developed a **multi-scale directed GNN learning treatment-aware transitions** across hepatocyte injury trajectories (36,067 cells); resolved temporally distinct recovery phases distinguishing genuine resolution from persistent compensation and transferred zero-shot across species and modalities to human liver-failure injury regions. Manuscript in preparation.
- **Undergraduate Thesis:** Investigated anxiolytic effects of *Indigofera bracteolata* using DeepLabCut-based pose estimation (>94% accuracy); quantified dose-dependent effects (Cohen’s $d=0.8$ –1.3 at 600 mg/kg) across three behavioral paradigms; automated pipeline reduced analysis time 6×. Paper under review at *Journal of Pharmacological and Toxicological Methods*.
- Examined structure–toxicity relationships in 30 NSAIDs via unsupervised chemoinformatics; found non-hepatotoxic NSAIDs cluster within hepatotoxic chemical neighborhoods, implicating mechanistic over structural determinants of DILI. [Code]

African Society for Bioinformatics & Computational Biology / NIH ODSS

Remote

Bioinformatics Researcher

Dec. 2025 – Present

- **absuite** — Developed an R/Bioconductor package (10 functions) for antibody repertoire profiling; validated on 3.4M sequences across COVID-19 and influenza cohorts, reproducing published convergent clonotypes at 98.8% concordance.

Accepted for presentation at **ISMB 2026** (oral), **BioC 2026** (oral), and **EuroBioC 2026** (poster). Package in preparation for Bioconductor submission.

Dept. of Pharmaceutical & Medicinal Chemistry, University of Nigeria, Nsukka

Nsukka, Nigeria

Undergraduate Researcher — Computational Drug Discovery

Sept. 2025 – Present

Supervisor: [Prof. Charles O. Nnadi](#)

- **CASPID** — Designed a context-aware neural conditioning framework across 23,248 EGFR/BRAF inhibitor–cell line pairs; learns context-dependent structural feature importance from cellular transcriptomic state, revealing interpretable MET-mediated bypass resistance patterns. *Manuscript in preparation for BMC Bioinformatics*. [\[Code\]](#)
- **TransQSAR-pf** — Developed a biology-informed QSAR framework integrating *P. falciparum* transcriptomic signatures with 125 compounds; 71.2% of predictive performance attributed to conserved stress-response genes over chemical descriptors alone. *Published in MDPI Engineering Proceedings*. [\[Paper\]](#) [\[Code\]](#) [\[Poster\]](#)

HackBio

Remote

Bioinformatics Research Intern — Next Generation Sequencing

Sept.–Nov. 2025

Supervisor: [Dr. Adewale Ogunleye, PhD](#)

- Performed guilt-by-association co-expression analysis on 69 HCC samples, identifying stage-specific lncRNAs tracking nivolumab+ipilimumab response, enriched in IFN/T-cell pathways, and validating a 12-lncRNA panel for monitoring immunotherapy progression. [\[Paper\]](#) [\[Code\]](#)

Genomac Institute Inc.

Remote

Cancer Genomics Research Intern

Jul.–Oct. 2025

- Performed multi-omics analysis of 123 meningioma samples (TCGA, cBioPortal), integrating somatic mutation and CNA data to validate TMB and CNA as candidate biomarkers for WHO-grade stratification. *Poster presented at ISPOR-HEOR UNN 2025 and CoGSAYR 2026*. [\[Code\]](#) [\[Poster\]](#) [\[Poster\]](#)

Person-Centered HIV Research Team (PeCHIVReT), University of Nigeria, Nsukka

Nsukka, Nigeria

Undergraduate Researcher — Clinical Epidemiology

Sept. 2024 – Present

Supervisor: [Dr. AbdulMuminu Isah, PharmD, PhD](#)

- Extracted and synthesized safety data from 113 HIV vaccine trials (1991–2024) for a systematic review; contributed to statistical analysis, evidence mapping, and manuscript preparation. *Paper under review at BMC Infectious Diseases*.

HackBio

Remote

Bioinformatics Intern — Cancer Genomics

Sept.–Nov. 2024

- Analyzed TCGA-BRCA and GBM RNA-Seq cohorts (1,200+ samples); identified three biomarkers (*CST5*, *CGA*, *MYL9*) with RF classification; GO/KEGG enrichment revealed biosynthetic dysregulation in gliomas. [\[Portfolio\]](#)

Publications

Bold = author; [†] equal contribution (co-first author); * corresponding author.

Peer-Reviewed & Conference Publications

- [1] **Igwezeke, F. O.**, Ugwuishiwu, C. H., Emesiani, J. U., Agada, S. I., Anozie, E. L., Kama, M. O., & Emegoakor, A. C. (2026). **MORPHA: Morphology-Constrained Training and the Limits of Cross-Acquisition Transfer in Low-Resource Malaria Microscopy**. *MIRASOL Workshop @ MICCAI 2026*. To appear in the *Lecture Notes in Computer Science* series (Springer).
- [2] Abubakar, J. I., Jarne, J., **Igwezeke, F.**, & Adewunmi, M. A. (2026). **CaresAI at SMM4H-HeaRD 2026: Predicting TNM Staging**. In *Proceedings of the 11th Workshop on Social Media Mining for Health (SMM4H-HeaRD) @ ACL 2026*.
- [3] Hamnett, L., **Igwezeke, F.**, Abubakar, J. I., & Adewunmi, M. A. (2026). **CaresAI at CT-DEB'26: Detecting Dosing Errors in Clinical Trials Using Domain-Specific Transformer Embeddings and Classification Models**. In *Proceedings of the Third Workshop on Patient-Oriented Language Processing (CL4Health 2026) @ LREC 2026*.
- [4] **Igwezeke, F. O.** & Nnadi, C. O. (2026). **TransQSAR-pf: A Bio-Informed QSAR Framework Using Plasmodium falciparum Stress Signatures for Enhanced Antiplasmodial Activity Prediction**. *Engineering Proceedings*, 124(1), 37.

Preprints & Manuscripts Under Review

- [5] **Igwezeke, F. O.**, Abdel-Salam, R., Idagu, E., & Adewunmi, M. (2026). CURE: Clinical Uncertainty-aware REflexive Graph-RAG for Multihop Biomedical Question Answering. *Under review at Machine Learning for Health (ML4H) 2026, Proceedings Track*.
- [6] **Igwezeke, F. O.** & Adewunmi, M. A. (2026). Mechanism or Memorisation? Stress-Testing Relational Generalisation in Drug–Target Interaction Prediction. *Under review at ICBINB-BIO: Failure Modes of AI in Biology, NeurIPS 2026 Workshop*.
- [7] Ogunleye, A. J.^{*}, El-Tony, Y.^{*}, Adebite, A. A.^{*}, **Igwezeke, F. O.**[†], Mahdi, S. M.[†], Mohamed, M. A.[†], Orimolade, T.[†], Osatoyinbo, O.[†] (2026). **Exploring lncRNAs as Biomarkers for Monitoring Hepatocellular Carcinoma Immunotherapy**

[Response](#). *Research Square*. Manuscript under review at *Gene*.

- [8] Onyeto, C. A., **Igwezeke, F. O.**, Uwakwe, C. C., et al. (2026). Deep Learning-Based Behavioural Analysis of Anxiolytic-Like Activity of *Indigofera bracteolata* Methanol Stem Extract in Wistar Rats. *Under review, Journal of Pharmacological and Toxicological Methods*.
- [9] Ayeni, V. A., Azumara, S. I., Agbo, C. E., **Igwezeke, F. O.**, et al. (2025). Safety Profile of HIV Vaccine Candidates Across Clinical Trial Phases: A Scoping Review. *Under review, BMC Infectious Diseases*.

Software

absuite — R/Bioconductor package for antibody repertoire profiling and clonotype analysis

10 analytical modules; validated on 3.4M BCR sequences across COVID-19 and influenza cohorts; in preparation for Bioconductor submission.

Selected Presentations

Oral Presentations

- [1] absuite: An R/Bioconductor Package for Antibody Repertoire Profiling and Clonotype Analysis. **BioC 2026** (Bioconductor Conference), Seattle, WA, Aug. 2026. *Short talk*. [\[Slides\]](#)
- [2] absuite: A Package in R for Antibody Repertoire Profiling and Clonotype Analysis. **ISMB 2026** (International Conference on Intelligent Systems for Molecular Biology), Jul. 2026. [\[Slides\]](#)
- [3] Anxiolytic Effects of *Indigofera bracteolata* Using Deep Learning-Based Automated Behavioral Analysis. *1st Int. Conference on Pharmaceutical, Herbal & Cosmetic Products*, UNN, Feb. 2026. [\[Slides\]](#)
- [4] Ensemble ML with Uncertainty Quantification for Pharmacogenetic-Guided Warfarin Dosing Across Diverse Populations. *CoGSAYR Africa Summit 2026*, Lagos, Jan. 2026.
- [5] Ensemble ML with Uncertainty Quantification for Pharmacogenetic-Guided Warfarin Dosing Across Diverse Populations. *ISPOR-HEOR UNN Conference*, Nov. 2025. 🏆 **2nd Place, Best Oral Presentation**. [\[Web App\]](#)
- [6] Unsupervised ML Framework for IDH Status Prediction in Low-Grade Gliomas. *5th Annual National Conference on Bioscience & Biotechnology*, UNN, Jul. 2025.
- [7] Exploratory Transcriptomic Analysis Reveals Preferential Downregulation of RNA Polymerase II in Recurrent Glioblastoma. *ISPOR-HEOR UNN Conference*, Feb. 2025. 🏆 **1st Place, Best Oral Presentation**. [\[Paper\]](#)

Poster Presentations

- [8] absuite: An R/Bioconductor Package for Antibody Repertoire Profiling and Clonotype Analysis. *EuroBioC2026*, Turku, Finland, 2026. [\[Poster\]](#)
- [9] Comparative Analysis of TMB and CNA in WHO-Graded Meningiomas. *CoGSAYR Africa Summit 2026*, Lagos, Jan. 2026. [\[Poster\]](#)
- [10] TransQSAR-pf: A Bio-Informed QSAR Framework Using *P. falciparum* Stress Signatures. *6th Int. Electronic Conference on Applied Sciences (ASEC 2025)*, p. 24, Dec. 2025. [\[Poster\]](#)
- [11] Transcriptomic Subtype Stratification of Low-Grade Gliomas by IDH Status. *NABIS 2025*, Ogbomoso, Oct. 2025. *NABIS 2025 Proceedings*, pp. 24–25.
- [12] Safety Profile of Candidate HIV Vaccines Across Clinical Trial Phases. *13th IAS Conference on HIV Science*, Kigali, Rwanda, Jul. 2025. *IAS 2025 Book of Abstracts*, p. 328.

Awards, Honours & Fellowships

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| Jun. 2026 | ISMB 2026 Conference Fellowship , International Society for Computational Biology (ISCB) — Competitive conference fellowship award (\$1,500 USD) |
| Jun. 2026 | BioC 2026 Travel Scholarship , Bioconductor Conference — \$550 USD travel award + complimentary registration |
| Mar. 2026 | Okwuchukwu Fund Research Award , UNN — NGN 120,000 competitive grant for undergraduate thesis research (1 of 3 recipients) |
| Dec. 2025 | ABI Travel Fellowship , African Bioinformatics Institute — Full support, CoGSAYR Africa Summit, Lagos |
| Nov. 2025 | 🏆 Gabriel Ezenri Award for Excellence in Research 2.0 — 2nd Place (NGN 110,000), ISPOR-HEOR UNN |
| Feb. 2025 | 🏆 Gabriel Ezenri Award for Excellence in Research — 1st Place (NGN 100,000), ISPOR-HEOR UNN |
| Feb. 2025 | 1st Runner-up , ISPOR-HEOR UNN Clinical Skills Evaluation Competition |
| Nov. 2024 | Research Fellow , Toufik's World Medical Association Research Fellowship in Neurology |
| Nov. 2024 | Udacity × Bertelsmann Next Generation Tech Booster Scholarship — Top 500 worldwide (\$1,200 USD) |
| Sept. 2024 | HackBio Precision Medicine Premium Scholarship — Selected 5 of 200+ applicants |
| Oct. 2023 | YandyTech × DataCamp Donates Data Science Scholarship (\$399 USD) |

Technical Skills

Programming	Python (PyTorch, TensorFlow, scikit-learn, RDKit), R (DESeq2, limma, Bioconductor), SQL, PySpark
Bioinformatics	TCGA/cBioPortal, GSEA, AlphaFold, PubChem API, TCGAbiolinks, AIRR/IMG/IT/OAS formats
ML/AI Methods	Transformers, GNNs, QSAR, Dimensionality Reduction (PCA, t-SNE), Clustering, Boruta
Cloud & DevOps	AWS, Docker, Git, HPC (Compute Ontario)

Professional Development

Jul. 2025	Cohere Labs ML Summer School — Transformers, diffusion models, self-supervised learning; faculty from Google DeepMind, Meta AI (FAIR), Adobe, INRIA, MIT. [Certificate]
Jun. 2025	Compute Ontario Summer School — HPC, bioinformatics pipelines (RNA-Seq), cloud computing, Docker/Git. [Certificate]